

ME 618: Stealth Technology: Infrared Signatures

Course Code	ME 618
Course Name	Stealth Technology: Infrared Signatures
L-T-P-C	2-1-0-3
Prerequisites	Heat Transfer (for those who Credit this course / no Pre-requisite for those who Audit this course)
Intended for	UG (ME final year) /MS (ME) /PhD (ME) can Credit this course [all other student categories can Audit this course (must at least pass in course to qualify as Audit)]
Distribution	Elective
Approval	5th Senate (Oct 2013)

Course Contents

- **Module I**
Principles of Stealth Technology – camouflage, conceal, deceive, active vs. passive detection. [2 Lectures]
- **Module II**
Introduction to Signatures – radar (RCS & its reduction), infrared, visual, aural. [3 Lectures]
- **Module III**
Principles of Thermal / Infrared Radiation – basic laws (Planck's, Wien's Displacement, Kirchhoff's), grey body spectrum. [6 Lectures]
- **Module IV**
Estimation of Infrared Signature (IR) from Internal Sources – engine heated casing, engine exhaust plume, aerodynamic heating of airframe. [6 Lectures]
- **Module V**
Estimation of Infrared Signature from External Sources – reflection of earthshine, sunshine, & skyshine. [6 Lectures]
- **Module VI**
Role of Atmosphere – attenuation of IR-signature by intervening atmosphere & atmospheric background radiance. [3 Lectures]
- **Module VII**
Relation between IR-Signature and Target Susceptibility – lock-on envelope & lethal envelop for air-to-air combat in horizontal plane. [6 Lectures]
- **Module VIII**
IR-Signature Suppression (& its Penalties) – optical blocking, cooling, emissivity optimization IR Countermeasures – decoys / flares [2 Lectures]

Text Books

1. Hudson Jr., R.D., **Infrared System Engineering**, Wiley Series in Pure and Applied Optics, 2006.
2. Hackforth, H.L., **Infrared Radiation**, McGraw Hill, 1960.
3. Jones, J., **Stealth Technology: The Art of Black Magic** (ed. Thurber, M.), McGraw-Hill Co. 1989.

References

1. Mahulikar, S.P., Sonawane, H.R., & Rao, G.A., 2007, **Infrared signature studies of aerospace vehicles**, *Progress in Aerospace Sciences*, Vol. 43, Nos. 7-8, pp. 218-245.
2. Rao, G.A., & Mahulikar, S.P., 2002 (Dec), **Integrated review of stealth technology and its role in airpower**, *Aeronautical Journal*, Vol. 106, No. 1066, pp. 629-641.
3. Howe D., **Introduction to the basic technology of stealth aircraft: Part 1 - basic considerations and aircraft self emitted signals (passive considerations)**, *ASME Journal of Engineering for Gas Turbines & Power*, 1991, Vol. 113, No. 75, pp. 75-79.